SAFE-CARE™ Framework: Executive Overview

A practical operating model for patient-facing healthcare AI agents

SAFE-CARE Framework · authored by Yassen Eltayeb, Conefia · Version 1.2.0 · July 2026

SAFE-CARE turns one order of priorities into an end-to-end operating model: recognize risk first, and sound helpful second. For example, when a patient says, “I have heavy bleeding and feel dizzy,” the agent must recognize risk, avoid reassurance, and route the user to prompt care.

The SAFE-CARE operating model

To do so, eight principles set practical requirements:

Principle	Practical requirement
S: Scope	Define intended use, prohibited actions, claims, users, and escalation boundary.
A: Anchor	Ground health claims in approved evidence and make support traceable.
F: Fence	Layer input, output, retrieval, tool, memory, and monitoring guardrails.
E: Evaluate	Use simulations, ablations, AI judges, human calibration, and regression tests.
C: Contextualize	Personalize from confirmed, time-stamped, correctable health context.
A: Act	Use permissioned tools with confirmation, audit logs, and rollback.

Principle	Practical requirement
R: Route	Escalate urgent, uncertain, severe, or regulated cases to humans.
E: Evidence	Maintain a safety case, registries, release report, and production monitoring.

In practice, the architecture is one chain: User → input safety → evidence/context → orchestrator → governed tools → output safety → response/citations → monitoring and human review.

Who benefits

On top of that, each stakeholder benefits differently:

Stakeholder	Value
Patients	Clear education, appropriate personalization, citations, and safe escalation.
Clinicians and operations	Lower repetitive follow-up burden while high-risk cases stay human-owned.
Product and engineering teams	A repeatable build-test-release pattern that replaces one-off prompt work.
Healthcare organizations and the field	A shareable safety baseline that raises the bar for patient-facing AI.

In one line

Healthcare AI safety is an operating system built around the model: scope, evidence, guardrails, tools, memory, evaluation, escalation, and governance.

Companion documents

These companion documents sit alongside this overview in the SAFE-CARE public release repository (github.com/Conefia/SAFE-CARE · DOI 10.5281/zenodo.21330841):

- **Menopause-support companion design case study**: the first applied case study of SAFE-CARE, based on the author's prior applied work and presented in anonymized form (docs/03_SAFE-CARE_Menopause-Support_Companion_Case_Study).
- **Building Safe Conversational AI Agents for Healthcare**: the conference talk outline (docs/09_SAFE-CARE_Conference_Talk_Deck_Outline).

Publication and citation

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